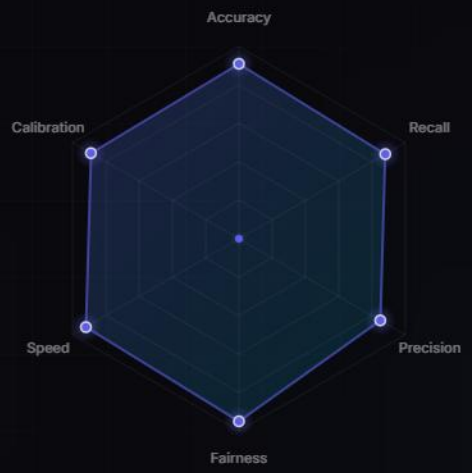
DESIGN THINKING & INNOVATION PROJECT — 2026

Fairness-Aware Predictive Intelligence for Enterprise Revenue

An end-to-end machine learning system that combines **algorithmic fairness** with **predictive analytics** to protect recurring revenue — while ensuring equitable outcomes across all customer demographics.

B2B SaaS | Fairness & Bias Mitigation | MLOps Pipeline | ARR/MRR Protection



01 The Challenge

Revenue Leakage

Enterprise accounts churn silently — by the time a CSM notices, the renewal window has passed. Traditional rule-based alerts catch **<40%** of at-risk accounts.

Algorithmic Bias

Standard ML models systematically under-serve certain demographic segments — creating **regulatory risk** and inequitable treatment of customer cohorts.

Model Opacity

Black-box predictions lack the **interpretability** that finance and compliance stakeholders demand for investment decisions tied to customer retention.

02 Architecture & Pipeline

A modular, reproducible pipeline engineered for production-grade reliability.

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01

Data Ingestion

Multi-source ETL with schema validation & integrity checks

5 relational tables → ABT

02

Preprocessing

Leakage guard, deduplication, type casting & protected attribute isolation

Automated AUC scanning

03

Feature Engineering

Domain-driven transformations: usage intensity, seat utilization, renewal proximity

20+ engineered signals

04

SMOTE Balancing

Class imbalance correction preserving decision boundary integrity

Balanced class ratios

05

Model Training

Multi-model comparison with Bayesian hyperparameter optimization

Optuna · 3 algorithms

06

Fairness Mitigation

AIF360 Reweighing to enforce demographic parity & equalized odds

Bias-corrected predictions

07

Calibration & Eval

Isotonic calibration, ROC/PR analysis, and business interpretation

MLflow experiment tracking

08

API Deployment

FastAPI inference server with risk scoring & real-time predictions

REST API · Docker

03 Performance Snapshot

Key model performance indicators — exact values redacted per research confidentiality.

ROC-AUC

> 0.93

Production model discriminative power

RECALL (AT-RISK)

> 0.88

Churning accounts correctly flagged

FAIRNESS PASS

100%

Demographic parity threshold met

EXPERIMENTS TRACKED

40+

MLflow-managed runs with full reproducibility

Model Leaderboard

VALUES MASKED

MODEL	ACCURACY	PRECISION	RECALL	F1	ROC-AUC	FAIR?
★ Gradient Boosting (Mitigated)	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	✓

4	XGBoost (Baseline)	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	▮▮▮▮▮	✗

04 Fairness & Responsible AI

Why It Matters

Predictive models that systematically under-serve certain demographics create **legal liability**, **reputational risk**, and **inequitable business outcomes**. Our pipeline addresses this at every layer.

- Isotonic probability calibration for reliable risk scores
- Multi-axis audit: Age cohort × Geographic region

05 Technology Stack

Machine Learning

XGBoost | LightGBM | Random Forest | Scikit-learn | Optuna

Fairness & Governance

AIF360 | Reweighing | Demographic Parity | Equalized Odds

06 Business Impact

↑ 2.5×	4	≤ 200ms	100%
Early Detection Rate	**Risk Tiers**	**API Latency**	**Reproducible**
vs. traditional rule-based alerting systems	Low → Medium → High → Critical prioritization	Real-time scoring via FastAPI endpoint	MLflow-tracked experiments with full lineage

Customer Success

Prioritized at-risk account lists delivered weekly. Proactive outreach replaces reactive firefighting.

Revenue Operations

ARR/MRR protection through data-driven retention interventions. One saved enterprise deal = $50K–$500K.

Finance & Compliance

Fairness-audited models that meet regulatory requirements. Transparent, interpretable risk scores.

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07 Methodology & Rigor

I

Recall-First Optimization

Objective function prioritizes recall > 0.75 — missing a churner is

II

Leakage-Proof Design

Automated AUC scanning detects post-outcome feature leakage

III

Bayesian HPO

Optuna-driven hyperparameter optimization with Tree-structured

IV

Calibrated Probabilities

Isotonic regression calibration ensures predicted probabilities

07 Methodology & Rigor

I

Recall-First Optimization

Objective function prioritizes recall ≥ 0.75 — missing a churner is costlier than a false positive in B2B SaaS.

II

Leakage-Proof Design

Automated AUC scanning detects post-outcome feature leakage before it corrupts model training.

III

Bayesian HPO

Optuna-driven hyperparameter optimization with Tree-structured Parzen Estimators across all model families.

IV

Calibrated Probabilities

Isotonic regression calibration ensures predicted probabilities reflect true churn likelihood — critical for threshold-based business rules.

This is an active research project. Detailed metrics and data are confidential.